

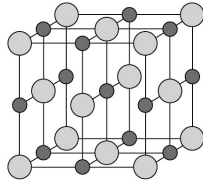
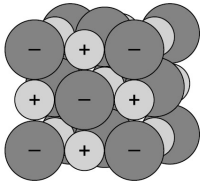
4.6.2 Structure and bonding

The theories of structure and bonding can explain the physical and chemical properties of materials. Analysis of structures shows that atoms can be arranged in a variety of ways, some of which are molecular while others are giant structures. Theories of bonding explain how atoms are held together in these structures. Materials scientists use this knowledge of structure and bonding to engineer new materials with desirable properties (see [Carbon chemistry](#) (page 137)).

4.6.2.1 Types of chemical bonding

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Describe and compare the nature and arrangement of chemical bonds in ionic compounds, simple molecules, giant covalent structures, and polymers and metals.	There are three types of strong chemical bonds: ionic, covalent and metallic. For ionic bonding the particles are oppositely charged ions. For covalent bonding the particles are atoms that share pairs of electrons. For metallic bonding the particles are atoms that share delocalised electrons. Ionic bonding occurs in compounds formed from metals combined with non-metals. Covalent bonding occurs in non-metallic elements and in compounds of non-metals. Metallic bonding occurs in metallic elements and alloys.	

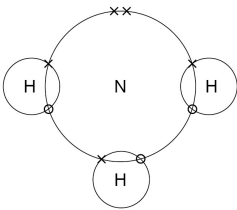
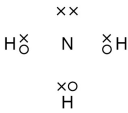
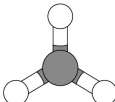
4.6.2.2 Ionic bonding

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain chemical bonding in terms of electrostatic forces and the transfer of electrons. Construct dot and cross diagrams for simple ionic substances. Deduce the empirical formula of a compound from the relative numbers of atoms present or from a model or diagram and vice versa. Use the formulae of common ions to deduce the formula of a compound.	When a metal atom reacts with a non-metal atom electrons in the outer shell of the metal atom are transferred. Metal atoms lose electrons to become positively charged ions. Non-metal atoms gain electrons to become negatively charged ions. The ions produced by metals in groups 1 and 2 and by non-metals in groups 6 and 7 have the electronic structure of a noble gas (Group 0). The electron transfer during the formation of an ionic compound can be represented by a dot and cross diagram, eg for sodium chloride: $\text{Na} \cdot + \cdot \ddot{\text{Cl}} \cdot \longrightarrow [\text{Na}]^+ [\cdot \ddot{\text{Cl}} \cdot]^-$ (2,8,1) (2,8,7) (2,8) (2,8,8) The charge on the ions produced by metals in groups 1 and 2 and by non-metals in groups 6 and 7 relates to the group number of the element in the periodic table. An ionic compound is a giant structure of ions. Ionic compounds are held together by strong electrostatic forces of attraction between oppositely charged ions. These forces act in all directions in the lattice and this is called ionic bonding. The structure of sodium chloride can be represented in the following forms:  Key ● Na⁺ ○ Cl⁻  Knowledge of the structures of specific ionic compounds other than sodium chloride is not required.	WS 1.2 Draw dot and cross diagrams for ionic compounds formed by metals in groups 1 and 2 with non-metals in groups 6 and 7. Work out the charge on the ions of metals and non-metals from the group number of the element, limited to the metals in groups 1 and 2, and non-metals in groups 6 and 7. Describe the limitations of particular representations and models to include dot and cross diagrams, ball and stick models and two- and three-dimensional representations. MS 4a Translate data between diagrammatic and numeric forms. MS 5b Draw or complete diagrams to represent 2D and 3D forms including two-dimensional representations of 3D structures. MS 1a Use arithmetic computation and ratio when determining empirical formulae.

4.6.2.3 Properties of ionic compounds

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the bulk properties of materials are related to the different types of bonds they contain, their bond strengths and the ways in which their bonds are arranged, recognising that the atoms themselves do not have these properties.	Ionic compounds have regular structures (giant ionic lattices) in which there are strong electrostatic forces of attraction in all directions between oppositely charged ions. These compounds have high melting points and high boiling points because of the large amounts of energy needed to break the many strong bonds. When melted or dissolved in water, ionic compounds conduct electricity because the ions are free to move and so charge can flow.	WS 1.2 Use ideas about energy transfers and the relative strength of chemical bonds and intermolecular forces to explain the different temperatures at which changes of state occur. Use data to predict states of substances under given conditions.

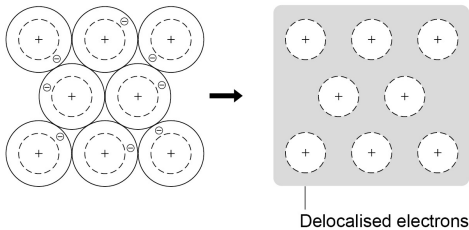
4.6.2.4 Covalent bonding

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain chemical bonding in terms of electrostatic forces and the sharing of electrons. Construct dot and cross diagrams for simple covalent substances. Deduce the empirical formula of a compound from the relative numbers of atoms present or from a model or diagram and vice versa.	When atoms share pairs of electrons they form covalent bonds. These bonds between atoms are strong. Covalently bonded substances may consist of small molecules. Some covalently bonded substances have very large molecules, such as polymers. Some covalently bonded substances have giant covalent structures, such as diamond and silicon dioxide. The covalent bonds in molecules and giant structures can be represented in the following forms: For ammonia (NH₃)  and/or  and/or  and/or  Polymers can be represented in the form: $\left(\begin{array}{cc} \text{H} & \text{H} \\ & \\ -\text{C} & - & \text{C}- \\ & \\ \text{H} & \text{H} \end{array} \right)_n$ poly(ethene) where n is a large number.	WS 1.2 Recognise substances as small molecules, polymers or giant structures from diagrams showing their bonding. Draw dot and cross diagrams for the molecules of hydrogen, chlorine, oxygen, nitrogen, hydrogen chloride, water, ammonia and methane. Represent the covalent bonds in small molecules, in the repeating units of polymers and in part of giant covalent structures, using a line to represent a single bond. Describe the limitations of particular representations and models to include dot and cross diagrams, ball and stick models and two- and three-dimensional representations. MS 5b Draw or complete diagrams to represent 2D and 3D forms including two-dimensional representations of 3D molecules. MS 1a Use arithmetic computation and ratio when determining

4.6.2.5 Properties of substances with covalent bonding

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the bulk properties of materials are related to the different types of bonds they contain, their bond strengths in relation to intermolecular forces and the ways in which their bonds are arranged, recognising that the atoms themselves do not have these properties.	Substances that consist of small molecules are usually gases or liquids that have relatively low melting points and boiling points. These substances have only weak forces between the molecules (intermolecular forces). It is these intermolecular forces that are overcome, not the covalent bonds, when the substance melts or boils. The intermolecular forces increase with the size of the molecules, so larger molecules have higher melting and boiling points. These substances do not conduct electricity because the molecules do not have an overall electric charge. Polymers have very large molecules. The atoms in the polymer molecules are linked to other atoms by strong covalent bonds. The intermolecular forces between polymer molecules are relatively strong and so these substances are solids at room temperature. Substances that consist of giant covalent structures are solids with very high melting points. All of the atoms in these structures are linked to other atoms by strong covalent bonds. These bonds must be overcome to melt or boil these substances. Diamond and graphite (forms of carbon) and silicon dioxide (silica) are examples of giant covalent structures.	WS 1.2 Use the idea that intermolecular forces are weak compared with covalent bonds to explain the bulk properties of molecular substances. Use ideas about energy transfers and the relative strength of chemical bonds and intermolecular forces to explain the different temperatures at which changes of state occur. Recognise polymers from diagrams showing their bonding. Recognise giant covalent structures from diagrams showing their bonding. Use data to predict states of substances under given conditions.

4.6.2.6 Metallic bonding

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain chemical bonding in terms of electrostatic forces and the sharing of electrons.	Metals consist of giant structures of atoms arranged in a regular pattern. The electrons in the outer shell of metal atoms are delocalised and so are free to move through the whole structure. The sharing of delocalised electrons gives rise to strong metallic bonds. The bonding in metals may be represented in the following form: 	MS 5b Draw or complete diagrams to represent 2D and 3D forms including two-dimensional representations of 3D structures.

4.6.2.7 Properties of metals

GCSE science subject content	Details of the science content	Scientific, practical and mathematical skills
Explain how the bulk properties of materials are related to the different types of bonds they contain, their bond strengths and the ways in which their bonds are arranged, recognising that the atoms themselves do not have these properties.	Metals have giant structures of atoms with strong metallic bonding. This means that most metals have high melting and boiling points. The layers of atoms in a metal crystal can slide over each other. This means metals can be bent and shaped. Pure metals are too soft for many uses and so are mixed with other metals to make alloys. The different sizes of atoms in an alloy distort the crystal structure, making alloys harder than pure metals. Metals are good conductors of electricity because the delocalised electrons in the metal carry electrical charge through the metal.	WS 1.2 Use ideas about energy transfers and the relative strength of chemical bonds and intermolecular forces to explain the different temperatures at which changes of state occur. Use data to predict states of substances under given conditions.